

Translated

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Gauss method for solving system of linear equations

Given a system of n linear algebraic equations (SLAE) with m unknowns. You are asked to solve the system: to determine if it has no solution, exactly one solution or infinite number of solutions. And in case it has at least one solution, find any of them.

Formally, the problem is formulated as follows: solve the system:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_m &= b_n \end{aligned}$$

where the coefficients a_{ij} (for i from 1 to n , j from 1 to m) and b_i (i from 1 to n) are known and variables x_i (i from 1 to m) are unknowns.

This problem also has a simple matrix representation:

$$Ax = b,$$

where A is a matrix of size $n \times m$ of coefficients a_{ij} and b is the column vector of size n .

It is worth noting that the method presented in this article can also be used to solve the equation modulo any number p , i.e.:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m &\equiv b_1 \pmod{p} \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m &\equiv b_2 \pmod{p} \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_m &\equiv b_n \pmod{p} \end{aligned}$$

Gauss

Strictly speaking, the method described below should be called "Gauss-Jordan", or Gauss-Jordan elimination, because it is a variation of the Gauss method, described by Jordan in 1887.